THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Patterns in Mathematics** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 21	Each term is the sum of the two terms before it.
2	**(a)** 3	Number of lines $= \frac{3\times 2}{2}$.
3	**(b)** 25	Two consecutive triangular numbers add up to a square number.
4	**(d)** 120	Triangular number $T_{15} = \frac{15\times 16}{2}$.
5	**(d)** 256	16×16.
6	**(d)** 45	Triangular numbers: term n is $1 + 2 + \cdots + n$.
7	**(a)** 216	Cube numbers are $1^3, 2^3, 3^3, \ldots$
8	**(a)** 34	Each term is the sum of the two terms before it.
9	**(b)** 6	Number of lines $= \frac{4\times 3}{2}$.
10	**(b)** 36	Two consecutive triangular numbers add up to a square number.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Each term is the sum of the previous two (1); $1, 1, 2, 3, 5, 8, 13, 21, 34, 55$ (1).	1+1
12	Each point joins to 7 others, and every line is counted twice (1); Lines $= \frac{8\times 7}{2} = 28$ (1).	1+1
13	Sum of first n odd numbers is $n \times n$ (1); $13 \times 13 = 169$ (1).	1+1
14	Each term is the sum of the previous two (1); $1, 3, 4, 7, 11, 18, 29, 47, 76, 123$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sum $= 45 + 55 = 100$ (1); $100 = 10 \times 10$ (1); Two consecutive triangular numbers always add to a square number (1).	1+1+1
16	$1 + 2 + \cdots + 3 = 6$ (1); $1^3 + \cdots + 3^3 = 1 + 8 + 27 = 36$ (1); $6^2 = 36$, so they match (1).	1+1+1
17	Difference between terms is 2 (1); 10th term $= 5 + 9 \times 2$ (add the difference 9 times) (1); $= 23$ (1).	1+1+1
	OR: Sum $= 55 + 66 = 121$ (1); $121 = 11 \times 11$ (1); Two consecutive triangular numbers always add to a square number (1).	1+1+1
18	$1 + 2 + \cdots + 4 = 10$ (1); $1^3 + \cdots + 4^3 = 1 + 8 + 27 + 64 = 100$ (1); $10^2 = 100$, so they match (1).	1+1+1
	OR: Difference between terms is 3 (1); 10th term $= 8 + 9 \times 3$ (add the difference 9 times) (1); $= 35$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	First person shakes 5 hands, next 4 new hands, ... down to 1 (1); Total $= 5 + 4 + 3 + 2 + 1$ (1); $= 15$ handshakes (1); This is the triangular number T_5 (2).	1+1+1+2
	OR: Next term $= 8 + 13 = 21$ (1); Then 34 and 55 (2); Sum of first 5 terms $= 19$ (equals the term two places ahead minus 2: 21-2) (2).	1+2+2
20	First person shakes 6 hands, next 5 new hands, ... down to 1 (1); Total $= 6 + 5 + 4 + 3 + 2 + 1$ (1); $= 21$ handshakes (1); This is the triangular number T_6 (2).	1+1+1+2
	OR: Next term $= 13 + 21 = 34$ (1); Then 55 and 89 (2); Sum of first 6 terms $= 32$ (equals the term two places ahead minus 2: 34-2) (2).	1+2+2

Mistakes to avoid:

- Thinking the sum of the first n natural numbers is $n \times n$; that is the sum of the first n **odd** numbers.
- Guessing a rule from only two terms. Always check the rule on at least three terms.
- Mixing up the nth odd number $2n - 1$ with the nth even number $2n$.
- Treating $2, 4, 8, 16$ as having a common difference; the gaps are $2, 4, 8$, so no constant is added.
- In the Virahanka sequence, adding the wrong two terms (the rule uses the *two terms just before*).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Patterns in Mathematics** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 210	Triangular number $T_{20} = \frac{20\times21}{2}$.
2	**(b)** 289	17×17.
3	**(a)** 78	Triangular numbers: term n is $1 + 2 + \cdots + n$.
4	**(c)** 343	Cube numbers are $1^3, 2^3, 3^3, \ldots$
5	**(d)** 55	Each term is the sum of the two terms before it.
6	**(c)** 10	Number of lines $= \frac{5\times4}{2}$.
7	**(c)** 49	Two consecutive triangular numbers add up to a square number.
8	**(a)** 325	Triangular number $T_{25} = \frac{25\times26}{2}$.
9	**(d)** 361	19×19.
10	**(b)** 91	Triangular numbers: term n is $1 + 2 + \cdots + n$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Each point joins to 8 others, and every line is counted twice (1); Lines $= \frac{9\times8}{2} = 36$ (1).	1+1
12	Sum of first n odd numbers is $n \times n$ (1); $15 \times 15 = 225$ (1).	1+1
13	Each term is the sum of the previous two (1); $2, 3, 5, 8, 13, 21, 34, 55, 89, 144$ (1).	1+1
14	Each point joins to 9 others, and every line is counted twice (1); Lines $= \frac{10\times9}{2} = 45$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sum $= 66 + 78 = 144$ (1); $144 = 12 \times 12$ (1); Two consecutive triangular numbers always add to a square number (1).	1+1+1
16	$1 + 2 + \cdots + 5 = 15$ (1); $1^3 + \cdots + 5^3 = 1 + 8 + 27 + 64 + 125 = 225$ (1); $15^2 = 225$, so they match (1).	1+1+1
17	Difference between terms is 4 (1); 10th term $= 11 + 9 \times 4$ (add the difference 9 times) (1); $= 47$ (1).	1+1+1
	OR: Sum $= 78 + 91 = 169$ (1); $169 = 13 \times 13$ (1); Two consecutive triangular numbers always add to a square number (1).	1+1+1
18	$1 + 2 + \cdots + 6 = 21$ (1); $1^3 + \cdots + 6^3 = 1 + 8 + 27 + 64 + 125 + 216 = 441$ (1); $21^2 = 441$, so they match (1).	1+1+1
	OR: Difference between terms is 9 (1); 10th term $= 1 + 9 \times 9$ (add the difference 9 times) (1); $= 82$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	First person shakes 7 hands, next 6 new hands, ... down to 1 (1); Total $= 7 + 6 + 5 + 4 + 3 + 2 + 1$ (1); $= 28$ handshakes (1); This is the triangular number T_7 (2).	1+1+1+2
	OR: Next term $= 21 + 34 = 55$ (1); Then 89 and 144 (2); Sum of first 7 terms $= 53$ (equals the term two places ahead minus 2: 55-2) (2).	1+2+2
20	First person shakes 8 hands, next 7 new hands, ... down to 1 (1); Total $= 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1$ (1); $= 36$ handshakes (1); This is the triangular number T_8 (2).	1+1+1+2
	OR: Next term $= 34 + 55 = 89$ (1); Then 144 and 233 (2); Sum of first 8 terms $= 87$ (equals the term two places ahead minus 2: 89-2) (2).	1+2+2

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- Mixing up the nth odd number $2n - 1$ with the nth even number $2n$.
- Treating $2, 4, 8, 16$ as having a common difference; the gaps are $2, 4, 8$, so no constant is added.
- In the Virahanka sequence, adding the wrong two terms (the rule uses the *two terms just before*).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Patterns in Mathematics** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 512	Cube numbers are $1^3, 2^3, 3^3, \ldots$
2	**(c)** 89	Each term is the sum of the two terms before it.
3	**(d)** 15	Number of lines $= \frac{6\times5}{2}$.
4	**(d)** 64	Two consecutive triangular numbers add up to a square number.
5	**(b)** 465	Triangular number $T_{30} = \frac{30\times31}{2}$.
6	**(d)** 441	21×21.
7	**(b)** 105	Triangular numbers: term n is $1 + 2 + \cdots + n$.
8	**(a)** 729	Cube numbers are $1^3, 2^3, 3^3, \ldots$
9	**(b)** 144	Each term is the sum of the two terms before it.
10	**(c)** 21	Number of lines $= \frac{7\times6}{2}$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sum of first n odd numbers is $n \times n$ (1); $17 \times 17 = 289$ (1).	1+1
12	Each term is the sum of the previous two (1); $1, 4, 5, 9, 14, 23, 37, 60, 97, 157$ (1).	1+1
13	Each point joins to 11 others, and every line is counted twice (1); Lines $= \frac{12\times11}{2} = 66$ (1).	1+1
14	Sum of first n odd numbers is $n \times n$ (1); $20 \times 20 = 400$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sum $= 91 + 105 = 196$ (1); $196 = 14 \times 14$ (1); Two consecutive triangular numbers always add to a square number (1).	1+1+1
16	$1 + 2 + \cdots + 7 = 28$ (1); $1^3 + \cdots + 7^3 = 1 + 8 + 27 + 64 + 125 + 216 + 343 = 784$ (1); $28^2 = 784$, so they match (1).	1+1+1
17	Difference between terms is 6 (1); 10th term $= 7 + 9 \times 6$ (add the difference 9 times) (1); $= 61$ (1).	1+1+1
	OR: Sum $= 105 + 120 = 225$ (1); $225 = 15 \times 15$ (1); Two consecutive triangular numbers always add to a square number (1).	1+1+1
18	$1 + 2 + \cdots + 8 = 36$ (1); $1^3 + \cdots + 8^3 = 1 + 8 + 27 + 64 + 125 + 216 + 343 + 512 = 1296$ (1); $36^2 = 1296$, so they match (1).	1+1+1
	OR: Difference between terms is 5 (1); 10th term $= 13 + 9 \times 5$ (add the difference 9 times) (1); $= 58$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	First person shakes 9 hands, next 8 new hands, ... down to 1 (1); Total $= 9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1$ (1); $= 45$ handshakes (1); This is the triangular number T_9 (2).	1+1+1+2
	OR: Next term $= 55 + 89 = 144$ (1); Then 233 and 377 (2); Sum of first 9 terms $= 142$ (equals the term two places ahead minus 2: 144-2) (2).	1+2+2
20	First person shakes 10 hands, next 9 new hands, ... down to 1 (1); Total $= 10 + 9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1$ (1); $= 55$ handshakes (1); This is the triangular number T_{10} (2).	1+1+1+2
	OR: Next term $= 89 + 144 = 233$ (1); Then 377 and 610 (2); Sum of first 10 terms $= 231$ (equals the term two places ahead minus 2: 233-2) (2).	1+2+2

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